SMA 2102: CALCULUS II CAT II

(May 12, 2026)

a. Evaluate $\int_{0.2}^{1} \frac{dx}{4-x^2}$ correct to 4 decimal places using trapezoidal rule with 8 subdivisions.

[8 Marks]

b. Evaluate the following integrals.

i) $\int \frac{dx}{x^2+6x+13}$ [6 Marks]

ii) $\int \frac{10x-2x^2}{(x-1)^2(x+3)}dx$ [6 Marks]

c. A company is trying to expose a new product to as many people as possible through television advertising. Suppose that the rate of exposure to the new people is proportional to the number of those who have not seen the product out of L possible viewers. No one was aware of the product at the start of the campaign and after 10 days 40% of L are aware of the product. Form a differential equation satisfied by the number N of people who will be aware of the product at any time t and state the condition N must satisfy. Hence determine what percentage of L will have been exposed after five days of the campaign. [10 Marks]
